

Supplementary Material

Adjusting to *Alice*: USPTO patent examination outcomes after *Alice Corp. v. CLS Bank International*

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April 2020

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Introduction

This supplement describes the data and research methods used in “Adjusting to Alice: USPTO patent examination outcomes after *Alice Corp. v. CLS Bank International*” published in April 2020.

Data construction and description

Database construction started with information on USPTO patent applications. That information was augmented in several ways to complete the datasets used in the empirical analysis. To evaluate our first examination outcome of interest – the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter – we estimated models using patent application-level datasets. For our second examination outcome – uncertainty in patent examination – we aggregated the application-level datasets to the technology class level.

Datasets on patent applications: public and proprietary versions

Two patent application-level datasets were constructed for the empirical analysis. A public data version, which was used to estimate models evaluating the U.S. Supreme Court decision in *Alice Corp. v. CLS Bank International* (*Alice*), and a proprietary data version, which was used to evaluate the USPTO’s Berkheimer Memorandum and the January 2019 Revised Patent Subject Matter Eligibility Guidance (2019 PEG). The proprietary dataset was required for the examiner guidance analysis to avoid problems with missing public data. Patent application information is not publicly available until about 18 months after initial filing and the time dimension for the analysis extends through the end of January 2020.

The public application datasets started with the Patent Examination Research Dataset (PatEx) described in Graham et al. 2018 (PatEx data are available at <https://www.uspto.gov/learning-and-resources/electronic-data-products/patent-examination-research-dataset-public-pair>), and the proprietary application dataset with the internal USPTO version of PatEx (PALM) that contains information on non-public applications. From these datasets, we extracted all the applications, granted or pending, with examiner office actions of the following types: non-final rejection, final rejection, ex parte quayle, and notice of allowance. The earliest mailing date shown in PatEx for an application was defined as the “first office action.”

The next step for the public dataset was to merge the patent applications from PatEx with the USPTO Office Action Research Dataset for Patents (OADP) described in Lu et al. 2017 (available at <https://www.uspto.gov/learning-and-resources/electronic-data-products/office-action-research-dataset-patents>). OADP provides comprehensive information on examiner issued rejections at all stages of prosecution. It covers office actions mailed during the period 2008 to mid-2017 and indicates whether the examiners’ office actions contain a non-final or final

rejection along with a description of the rejection types. The rejection type of interest is 35 USC §101 (Section 101) as it indicates a rejection for patent-ineligible subject matter. Note that each office action may have multiple rejections and rejection types. OADP was merged with PatEx using the patent application numbers and the office action mailing dates.

For the proprietary dataset, we merged a proprietary version of the OADP to obtain information on office actions for non-public applications. For both public and private applications, PALM identified the examiner who issued the first office action. PatEx only identifies the last examiner associated with the patent application and, in many cases, that person is not the one who issued the first office action.

Description of patent application characteristics

We added several characteristics to the application-level dataset. For each application, when available, we added the U.S. Patent Classification (USPC) listed on the pre-grant publication (PGPub). The classification listed on the PGPub is most likely the classification at the time of the examiner's first office action. For granted patents without a PGPub, we used the USPC listed on the granted patents. For a small number of applications that were published after USPTO discontinued the USPC system in 2015, we had to use the most recent USPC from PatEx. Lastly, for unpublished applications, we used the latest classification as listed in the PALM system.

For the public dataset, we augmented the application information with text-based data elements computed from published and granted patents. From the Patent Claims Research Dataset described in Marco et al. 2019 we added a variable to capture claim scope (the "breadth" of a claimed invention) by using the independent claim count (ICC) and shortest independent claim (ICL). (That dataset is available at <https://www.uspto.gov/learning-and-resources/electronic-data-products/patent-claims-research-dataset>.) Next, we developed a text-based measure of abstract language in patent claims (details available upon request). The source for patent application claims text is the USPTO bulk XML data (<https://bulkdata.uspto.gov/>). For each patent application, we used the text in the PGPub (when available) or the text in the granted patent. Because these text-based data elements rely on publicly available information, they are not available for our proprietary dataset used to evaluate the Berkheimer Memorandum and the 2019 PEG.

The public and proprietary patent application-level datasets are pooled cross-sections of applications grouped into one month cohorts by filing date (i.e., application-filing month observations). These datasets were used to estimate the models for the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter.

Datasets at the technology class level

To evaluate the degree of uncertainty in the patent examination process, we constructed two new datasets. In these datasets, the application-level observations were aggregated to broader technology classes (3-digit USPCs) and the time dimension was expanded to cover six month intervals (i.e., technology-half year observations). These adjustments were necessary in order to measure the degree of variation across examiners in their decision-making outcomes. The decision outcomes, which are described in detail in the next section, are each examiner's proportion of first office actions with rejections for patent-ineligible subject matter. We calculated the variance across examiners in their decision outcomes within a technology area and time period (i.e., the variance across examiners within technology-half year cells).

For the other data elements, when those are available in the public and private application datasets, we used aggregated values calculated by taking their average for the technology-half year cells. For instance, for counts of independent claims in the applications, the values are the average number of independent claims among applications filed in a particular 3-digit USPC and specific half-year time period. We also added a new data element: the average of the variance across examiners in their decision outcomes. When included in the models, this "level of variance" in the technology-half year cells helps to identify that portion of changes in uncertainty attributable to examiner decisions about the event under study such as the *Alice* decision. Changes in uncertainty within USPC 3-digit technology areas might reflect examiner compositional changes such as the adding of new examiners with less experience or changes in how applications are assigned to examiners. It would be wrong to attribute this type of influence to the event under study. Some other new data elements include the number of examiners and the number of applications per technology-half year.

The public and proprietary technology-half year datasets are panels with multiple observations for each technology class. These datasets were used to estimate the models for examination uncertainty in the first office action stage of patent examination.

Restrict to applications filed before *Alice*, Berkheimer memorandum and USPTO 2019 PEG

A potential source of endogeneity for identifying the impact of *Alice* and subsequent guidance documents on examiner outcomes is applicant behavior. In particular, applicants may react to the *Alice* decision and guidance documents by submitting redrafted applications, or prosecuting applications differently after the events. To alleviate these endogeneity concerns, we restricted our datasets to applications filed before the relevant event. This restriction mitigates the influence of applicant drafting strategies on our results. In a robustness check, we explored the implications of this restriction.

Additionally, our outcomes of interest are at the first office action stage of patent examination. Subsequent examination stages involve additional decision makers such as attorneys, agents, supervisors, and so forth. Thus, using a later stage of patent examination would confound the decisions of examiners with many others. Further, the first office action describes the initial patentability decision by the examiner and is performed with minimal interaction with applicants, which further reduces the influence of applicant behavior on the examination outcomes in this study.

Restriction to adjust for the Supreme Court Case *Bilski v. Kappos*

In 2010, the U.S. Supreme Court decided *Bilski v. Kappos* (*Bilski*), which was another case involving a business method patent. The Court found the claims to be patent-ineligible because they were directed toward an abstract idea. For this analysis, the *Bilski* decision may affect the pre-event trends (e.g., the trends before the *Alice* decision) in the *Alice*-affected technologies and other technologies. As discussed below, the empirical design of this study requires a statistically indistinguishable pre-event trend in *Alice*-affected and other technologies. To address this potential issue, the regression dataset starts in 2013 instead of 2011, thereby minimizing any impact from the *Bilski* decision. In a robustness check of the results, we relaxed this restriction and used the data from 2011. The results are quantitatively very similar and qualitatively the same.

Definition of “*Alice*-affected” and other technology groups

To identify the technologies affected by *Alice* (called *Alice*-affected technologies), we drew on patent litigation cases where validity was challenged under the “abstract idea” judicial exception. The USPTO’s Office of Patent Legal Administration (OPLA) maintains a public and comprehensive list of cases from the U.S. Court of Appeals for the Federal Circuit (CAFC) and the U.S. Supreme Court. A team of lawyers monitors these cases and identifies patents litigated for subject matter eligibility. One of the fields recorded for each case is the “Exception Type.” This refers to the relevant judicial exception type (i.e. laws of nature, products of nature, natural phenomena, or abstract ideas).

For this analysis, we defined *Alice*-affected technologies as USPCs appearing in cases involving the abstract ideas judicial exception. This group of technologies has 33 USPCs and comprises 33% of the patent applications in our dataset. The group of other technologies includes all USPCs that were not involved in any decision from the cases considered by OPLA and contains applications from 382 different USPCs.

Outcome variable definitions

Percentage of first office action Section 101 rejections

Our first outcome of interest is the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter (also referred to as a Section 101 rejection). This is a binary indicator that takes the value of 1 when the first office action includes a rejection for patent-ineligible subject matter, 0 otherwise. The binary indicator is the dependent variable in the regression models, but the descriptive figures shown in the *Adjusting to Alice* report aggregated this indicator by technology and month. Specifically, those figures show the percentage of first office action Section 101 rejections among all first office actions in the technology groups of interest, *Alice*-affected and other technologies, for a given month.

Section 101 first action examination uncertainty

The second outcome measure captures one source of uncertainty in the patent examination process. As described above, the analysis focused on uncertainty surrounding examiners' decisions in the first action stage of patent examination. The uncertainty metric was computed in two steps. First, within technology-half year cells, the first office actions for each application were aggregated to the examiner-level. This allowed us to calculate the share of first office action Section 101 rejections for each examiner. Next, the variance across the examiners' shares was used to capture uncertainty. This metric will increase when examiners' determinations on subject matter eligibility are more uneven within a technology-half year.

For our descriptive analysis in the *Adjusting to Alice* report, Section 101 first office action examination uncertainty is the average of the variance observed in each technology-half year for the groups of interest, *Alice*-affected and other technologies.

Methodology

Alice effect on the probability of first office action Section 101 rejections

To identify the impact of *Alice* on the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter, we used a difference-in-differences methodology with a linear probability model. The baseline estimating equation (see Table 2 in the Findings Section) is:

$$pr(101)_{it} = \beta_0 + \beta_1(PatGrp_{it} * Alice_t) + \delta_k + \theta_j + \gamma_t + \alpha X_{it} + \epsilon_{it} \quad (1)$$

where i indicates the patent application, j is the examiner, k is the technology area (3-digit USPC), t is time, $pr(101)_{it}$ is a binary variable that indicates whether the application received a Section 101 rejection on the first action, δ_k is a technology fixed effect, γ_t is a time fixed effect,

θ_j is an examiner fixed effect, and $\mathbf{X}_{it}$ is a vector of patent characteristics. *Alice*-affected technologies are represented by the variable, $PatGrp_{it}$, and the event indicator for the *Alice* decision is captured by $Alice_t$. The event indicator takes the value of 0 before June 2014 and the value 1 thereafter. We used robust standard errors to account for heteroskedasticity. The coefficient, β_1 , is the “treatment” effect. The magnitude and statistical significance of this coefficient reveals how the *Alice* decision changed the probability of receiving a first office action with a Section 101 rejection. The impact of the *Alice* decision is evaluated for those technologies exposed to the *Alice* standard (i.e., *Alice*-affected technologies in $PatGrp_{it}$) relative to the other technologies. This relative comparison eliminates common influences affecting all technologies such as an increase in Section 101 rejections due to utility and statutory double patenting.

Additional co-variates for non-random assignment

Even though the *Alice* decision is plausibly exogenous to the USPTO and its examiners, the assignment of patent applications to the *Alice*-affected group is not random because it relies on litigation events and the technologies of the patents involved. The use of OPLA’s information on litigation cases resulted in a broad definition of technologies potentially impacted by *Alice*. This is a conservative way to define the “treatment” group for the analysis. However, the model specifications also include a variety of patent application characteristics, $\mathbf{X}_{it}$, such as claim scope to help alleviate concerns of possible omitted co-variates. Due to the nature of the *Alice* decision, we also explored whether the degree of abstractness contained in the language of the patent application was a confounding influence. We estimated a specification that includes the abstractness variable $Abst_i$ for application i as follows (see Table 2 in the Findings Section):

$$pr(101)_{it} = \beta_0 + \beta_1(PatGrp_{it} * Alice_t) + \beta_2 Abst_i + \delta_k + \theta_j + \gamma_t + \alpha \mathbf{X}_{it} + \epsilon_{it} \quad (2)$$

Differential *Alice* impact due to abstract language

We also explored the possibility that the *Alice* decision impacted *Alice*-affected technologies differentially according to the degree of abstract language contained in the patent applications. To do this we added the following interaction term to the model: $(PatGrp_{it} * Abst_i * Alice_t)$. The empirical specification becomes (see Table 2 in the Findings Section):

$$pr(101)_{it} = \beta_0 + \beta_1(PatGrp_{it} * Alice_t) + \beta_2 Abst_i + \beta_3(PatGrp_{it} * Abst_i * Alice_t) + \delta_k + \theta_j + \gamma_t + \mathbf{X}_{it}\alpha + \epsilon_{it} \quad (3)$$

With this model, the impact of the *Alice* decision or “treatment effect” is: $\beta_1 + \beta_3 * Abst_i$. If β_3 is positive and statistically significant, then the *Alice* decision caused an additional increase in the likelihood of receiving Section 101 for those applications containing more abstract language in the group of *Alice*-affected technologies.

Assessment of pre-trends

One of the key assumptions required for identifying causal effects with a difference-in-difference research design is that the pre-event trends in *Alice*-affected and other technologies are the same (or at least statistically indistinguishable). To verify this assumption, we estimated a model that interacts an indicator variable for *Alice*-affected technologies, $PatGrp_{it}$, and a monthly time indicator, $Month_t$, that begins in January 2013 and runs through December of 2016. For the pre-event trends assumption to hold, the coefficient estimates on the interaction terms should not be statistically different from zero before the *Alice* decision and should be statistically different after the *Alice* decision. The estimation equation is given by (see Figure 1 in the Findings Section):

$$pr(101)_{it} = \beta_0 + \sum_t^{2013-2016} \beta_t (PatGrp_{it} * Month_t) + \psi_1 Abst_i + \delta_k + \theta_j + X_{it}\alpha + \epsilon_{it} \quad (4)$$

Berkheimer memo and 2019 PEG effects on the probability of first office action Section 101 rejections

To assess the effects of the Berkheimer memorandum and the 2019 PEG on the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter, we used the same modeling approach described for the *Alice* decision. The event indicator for the Berkheimer memorandum takes the value of 0 before April 2018, and 1 thereafter. For the 2019 PEG, the event indicator takes the value of 0 before January 2019, and 1 thereafter. These event indicators are interacted with the *Alice*-affected technologies. However, the model specifications for these two events are more parsimonious. Due to the limitations of the proprietary patent application dataset, the specifications did not include application-level co-variates such as claim scope and application language abstractness.

Alice effect on first action examination uncertainty

To identify the impact of *Alice* on first action examination uncertainty, we use the difference-in-difference model given by (see Table 3 in the Findings Section):

$$Var_{kt} = \beta_0 + \beta_1 (PatGrp_{kt} * Alice_t) + \delta_k + \gamma_m + X_{kt}\alpha + \epsilon_{kt} \quad (5)$$

where k indicates the technology area, t is time in half-year intervals, δ_k is a technology (USPC) fixed effect, γ_t is a half-year fixed effect, $Alice_t$ is an event indicator variable taking a value of 1 after the *Alice* decision, and X_{kt} is a vector of USPC-level by time period co-variates. $PatGrp_{it}$ indicates whether the application is part of the *Alice*-affected technologies treatment group. The dependent variable Var_{kt} is the variance of Section 101 rejection rates across examiners in technology k in period t . That variable measures the decision-making spread in the

determinations of Section 101 rejections by examiners and, all else equal, greater variation reflects greater uncertainty about whether the patent application will receive a rejection for patent-ineligible subject matter in the first action stage of examination.

In the specification shown in eq. (5), we included a variety of patent application characteristics X_{it} averaged across the applications within each technology-half year cell. Besides the patent claim scope co-variables used in the models for Section 101 rejections, we added the variance in the abstractness of the language in the applications that examiners' received within each technology-half year cell. If examiners were assigned applications with varying degrees of abstract language after the *Alice* decision, Section 101 rejection decision variance may increase mechanically rather than reflect decision-making uncertainty from *Alice*. The abstractness co-variate addresses this potential issue.

We added several more co-variables to capture other sources of increased variance that may not be attributable to the *Alice* decision. These include within-examiner Section 101 rejection rate variances (the standard errors of each examiner's Section 101 rejection rate affects the variance across examiners), the number of examiners, and applications per examiner within each USPC at time t (because the number of examiners and applications affects the variance across examiners). These controls may be important as our dataset restricts the applications to those filed before *Alice*, which was done as a way to mitigate any influence from applicant behavior. This dataset restriction means that the number of first office actions will decrease as time passes because more and more of those applications have received a first office action decision. Such a mechanistic decrease could lead to an increase in the variance across examiners that does not reflect *Alice*.

Additional co-variables for non-random assignment

Similar to the specification for Section 101 rejections, we estimated a second model that includes the average abstractness measure, Avg_Abst_i . This co-variate is the average abstractness of the patent applications receiving a first office action in the USPC at time t . The coefficient will capture any direct effect of abstract language on examination uncertainty.

Differential *Alice* impact due to abstract language

We also explored the possibility that the *Alice* decision impacted first action examination uncertainty differentially according to the degree of abstract language contained in the patent applications. The approach is the same as discussed above for Section 101 rejections. Specifically, in a third model, we added a triple interaction term.

Assessment of pre-trends

To verify that the pre-event trends in examination uncertainty for *Alice*-affected and other technologies are statistically indistinguishable, we estimated a model that interacts an indicator

variable for *Alice*-affected technologies, $PatGrp_{it}$, with a half-year time indicator, $HalfYear_t$, that begins in January 2013 and runs through December of 2016. For the pre-event trends assumption to hold, the coefficient estimates on the interaction terms should not be statistically different from zero before the *Alice* decision. The estimation equation is given by (see Figure 2 in the Findings Section):

$$Var_{kt} = \beta_0 + \sum_t^{2013-2016} \beta_t (PatGrp_{it} * HalfYear_t) + \psi_1 Avg_Abst_i + \delta_k + \theta_j + X_{it}\alpha + \epsilon_{it} \quad (6)$$

Berkheimer memo and 2019 PEG effects on first action examination uncertainty

Once again, we used a similar approach to estimate the effects of the Berkheimer memorandum and the 2019 PEG on first action examination uncertainty. The event indicators are defined as before and included in the regression models. There are data limitations from using the proprietary dataset. For instance, we could not include several application characteristics such as average abstractness, average claim scope (both ICL and ICC), and the variance in abstractness across examiners. Despite this, we were able to include fixed effects, the number of applications, and the number of applications per examiner to alleviate concerns related to our identifying assumptions.

Findings

Descriptive analysis

The figures that descriptively illustrate the findings are presented the *Adjusting to Alice* report. Table 1 provides the mean values and standard deviations for three groups – other technologies, *Alice*-affected technologies, and combined – for the public dataset, before and after the *Alice* decision. *Alice*-affected technologies have a higher degree of abstractness compared to other technologies, but the *Alice* decision did not change the degree of abstractness. For instance, abstractness for *Alice*-affected technologies did not change much after *Alice* (0.02416 to 0.02419). This is desired and reflects the dataset restriction to only those applications filed before the *Alice* decision. The outcomes of interest – the percentage of first office actions with a Section 101 rejection and uncertainty in first action examination – both increased for *Alice*-affected and other technologies; however, the increase in the *Alice*-affected technologies is substantially larger. These differences anticipate the regression findings presented below.

Table 1: Summary Statistics

	Other technologies		<i>Alice</i> -affected technologies		Combined	
	<i>mean</i>	<i>std dev</i>	<i>mean</i>	<i>std dev</i>	<i>mean</i>	<i>std dev</i>
Application Abstractness	0.01679	0.0093403	0.02418	0.012319	0.01923	0.01098
Application Abstractness [before <i>Alice</i>]	0.01682	0.0093729	0.02416	0.012343	0.01918	0.01097
Application Abstractness [after <i>Alice</i>]	0.01676	0.0093095	0.02419	0.0123	0.01927	0.011
Percent of § 101 Rejections [before <i>Alice</i>]	5.42%	22.63%	21.11%	40.81%	10.45%	30.60%
Percent of § 101 Rejections [after <i>Alice</i>]	6.18%	24.08%	28.34%	45.07%	13.68%	34.37%
§ 101 Examination Variance [before <i>Alice</i>]	0.02665	0.0454264	0.09105	0.044542	0.03204	0.04872
§ 101 Examination Variance [after <i>Alice</i>]	0.02984	0.0547089	0.12266	0.065881	0.03808	0.06171

Regression results

Alice effect on first office action Section 101 rejections

Table 2 presents the main regression findings for the impact of the *Alice* decision on the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter. Model 1 shows the coefficient estimates for the baseline specification (eq. 1). The coefficient, β_1 , on the interaction term (*Alice* * *PatGrp_{it}*) is 0.0466, and is statistically significant at the 1% level. The result shows that the *Alice* decision increased the probability of receiving a Section 101 rejection in the first office action by 4.66 percentage points for a patent application in *Alice*-affected technologies.

Table 2: Regression Results for the *Alice* Decision on Section 101 Rejections

VARIABLES	Model 1	Model 2	Model 3
Abst		0.528*** (0.0341)	0.186*** (0.0362)
Alice x PatGrp	0.0466*** (0.00148)	0.0467*** (0.00148)	0.0140*** (0.00236)
Alice x PatGrp x Abst			1.364*** (0.0837)
ICC	0.00927*** (0.000284)	0.00915*** (0.000283)	0.00918*** (0.000284)
ICL	-7.18e-05*** (3.73e-06)	-7.77e-05*** (3.83e-06)	-7.57e-05*** (3.82e-06)
Constant	0.0909*** (0.00835)	0.0830*** (0.00836)	0.0879*** (0.00836)
Observations	1,027,924	1,027,924	1,027,924
R-squared	0.308	0.309	0.309
Time FE	Month x Year	Month x Year	Month x Year
Technology FE	USPC	USPC	USPC
Examiner FE	Yes	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Model 2 contains the application-level abstractness measure as in eq. (2). The coefficient on our main variable of interest, (*Alice* x PatGrp), remains unchanged. The estimate for the abstractness term is 0.528, indicating that a one standard deviation increase in the abstractness measure leads to a .58 percentage point increase in the probability of receiving a § 101 rejection.¹

Model 3 reports the results from the specification that contains the triple interaction term between the treatment group, the abstractness measure, and the *Alice* event indicator (eq. 3). The triple interaction term is statistically significant with a coefficient of 1.364. Using the standard deviation from the after-*Alice* abstractness for applications in *Alice*-affected technologies from Table 1 (i.e., 0.0123), the estimate indicates that the *Alice* decision had a

¹ This is calculated using the coefficient estimate from Table 2 and the standard deviation of the abstractness measure for all USPCs in Table 1, (0.01098*0.528).

stronger effect on applications with more abstract language. The probability of receiving a first office action with a rejection for patent-ineligible subject matter increased by 1.7 percentage points (i.e., 1.364×0.0123) for an application with one standard deviation higher degree of abstract language.

Note that the coefficient on (*Alice* x PatGrp) remains statistically significant at the 1% level with a coefficient estimate of 0.014. With both interaction terms positive and significant, the evidence suggests that the *Alice* decision had its impact on *Alice*-affected technologies through two channels, a direct effect on the *Alice*-affected technologies and a further effect when these patent applications contained more abstract language. The combined effect from the *Alice* decision, when evaluated at the mean abstraction in the *Alice*-affected group, is 4.7 percentage points ($\beta_1 + \beta_2 \times .0241896$) and is significant at the 1% level.

Figure 1 displays the point estimates for the interactions terms between *Alice*-affected technologies and the monthly time indicators, ($PatGrp_{it} \times Month_t$). The base is the month immediately before the *Alice* decision, May 2014. An F-test for the null hypotheses that all the pre-*Alice* interaction terms are significantly different from zero is rejected at the 1% level. Two months, September and October 2013, differ slightly while all the other months are not statistically different from zero using 95% confidence interval. This evidence supports the pre-trends assumption for the difference-in-difference design. Also apparent is a sharp increase in the probability of a Section 101 rejection in the period immediately following *Alice* (July 2014). Using this model, relative to the other technologies, the probability of receiving a first office action with a Section 101 rejection increased by 31% through December 2015.²

² This is computed in the following way. Let t_1 be the average probability of receiving a Section 101 rejection in the treatment group in May 2014, and t_2 be the average probability of receiving a Section 101 rejection in the treatment group in December 2015. c_1 and c_2 are the averages for the control group in these two periods. Then, 31 percent is $\frac{(t_2 - t_1) - (c_2 - c_1)}{t_1} \times 100$.

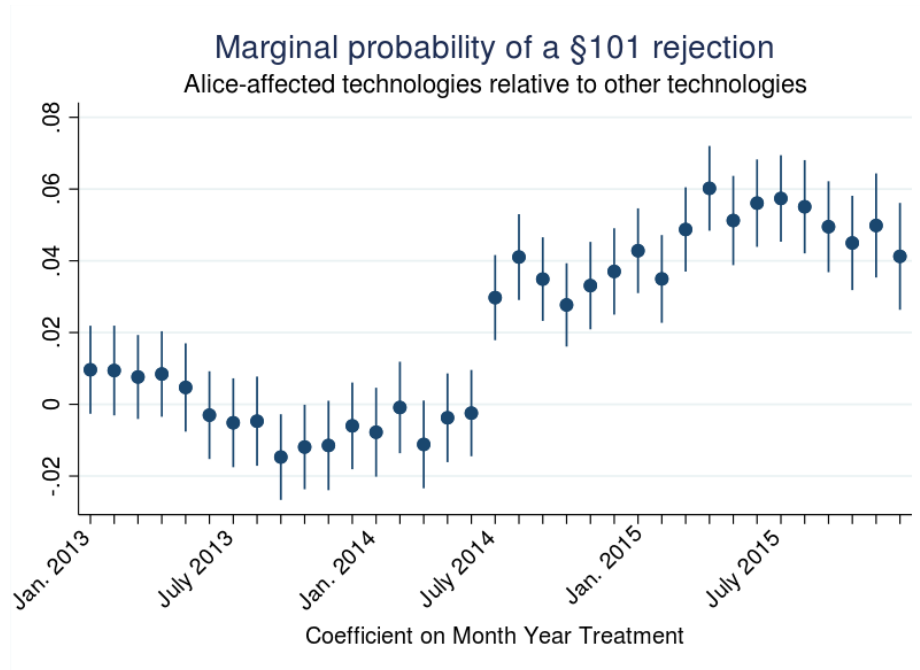


Figure 1: Monthly estimates for the marginal increase in Section 101 rejection rates in *Alice*-affected technologies relative to the control technologies (the base period is May 2014). The figure shows a clear change post-*Alice*, and that the pre-trend assumption of constant pre-trends is satisfied for one and a half years before *Alice*.

Berkheimer memo and 2019 PEG effects on first office action Section 101 rejections

Table 3 shows the regression results for the effects of the Berkheimer memorandum and the 2019 PEG on the likelihood of receiving a first office action with a rejection for patent-ineligible subject matter. For comparison, Model 1 in column (2) presents the results previously discussed in Table 2 for the *Alice* decision. In the next column, Model 5 shows that the Berkheimer memorandum reduced the likelihood of receiving a first office action Section 101 rejection for *Alice*-affected technologies. The estimate is -0.0157 and is statistically significant at the 1% level. It indicates the Berkheimer memo reduced the probability by 1.57 percentage points.

Model 6 shows that the 2019 PEG is associated with a larger decrease in the probability of receiving a first office action with a rejection for patent-ineligible subject matter. The coefficient estimate is over seven times larger at -0.116 and is significant at the 1% level. Using a model similar to eq. 4, the 2019 PEG reduced the likelihood of receiving a first office action with a Section 101 rejection by 25% for *Alice*-affected technologies between Dec. 2019 and Jan. 2020.³

³ Although equation (4) contains application level co-variables, for reasons described earlier, we do not use these independent variables for this estimate.

Table 3: Regression Results for the Berkheimer Memo and 2019 PEG on Section 101 Rejections

VARIABLES	Model 1	Model 4	Model 5
<i>Alice</i> x PatGrp	0.0466*** (0.00148)		
Berkheimer x PatGrp		-0.0157*** (0.00179)	
2019 PEG x PatGrp			-0.116*** (0.00144)
ICC	0.00927*** (0.000284)		
ICL	-7.18e-05*** (3.73e-06)		
Constant	0.0909*** (0.00835)	0.137*** (0.0117)	0.137*** (0.00992)
Observations	1,027,924	703,222	1,047,734
R-squared	0.308	0.439	0.387
Time FE	Month x Year	Month x Year	Month x Year
Technology FE	USPC	USPC	USPC
Examiner FE	Yes	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

***Alice* effect on first action examination uncertainty**

Table 4 presents the results for the effect of the *Alice* decision on first action examination uncertainty. In Model 6 the variable of interest is the interaction term between the *Alice* decision indicator and the group of *Alice*-affected technologies from eq. 5, *Alice* x PatGrp. The coefficient estimate is positive and statistically significant at the 1% level. For *Alice*-affected technologies, the size of the estimate indicates that the variance across examiners in Section 101 determinations increased by roughly 61% using the standard deviation observed before the *Alice* decision (information from Table 1, 61%=0.0272/0.044542*100).

Table 4: Regression Results for the *Alice* Decision on Examination Uncertainty

VARIABLES	Model 6	Model 7	Model 8
Avg. abstractness		-1.446 (0.946)	-1.458 (0.894)
<i>Alice</i> x PatGrp	0.0272*** (0.00453)	0.0272*** (0.00447)	0.0218 (0.0473)
<i>Alice</i> x PatGrp x Avg. abstractness			0.230 (1.947)
Avg. number of independent claims	-0.00409 (0.00465)	-0.00329 (0.00450)	-0.00325 (0.00438)
Avg. word count in claims	-9.23e-05 (8.29e-05)	-7.50e-05 (8.47e-05)	-7.45e-05 (8.39e-05)
Avg. examiner variance	0.144* (0.0826)	0.148* (0.0809)	0.150* (0.0837)
Number of examiners	-0.000105*** (3.32e-05)	-0.000105*** (3.29e-05)	-0.000105*** (3.32e-05)
Number of applications	8.36e-06** (3.77e-06)	8.45e-06** (3.77e-06)	8.48e-06** (3.76e-06)
Avg. applications per examiner	-0.00350*** (0.000780)	-0.00348*** (0.000775)	-0.00347*** (0.000776)
Variance in abstractness	-43.60 (54.07)	-20.96 (49.25)	-21.08 (49.18)
Constant	0.0791*** (0.0179)	0.0977*** (0.0206)	0.0978*** (0.0206)
Observations	3,030	3,030	3,030
R-squared	0.551	0.552	0.552
Time FE	Half Year	Half Year	Half Year
Technology FE	USPC	USPC	USPC

Robust standard errors in parentheses

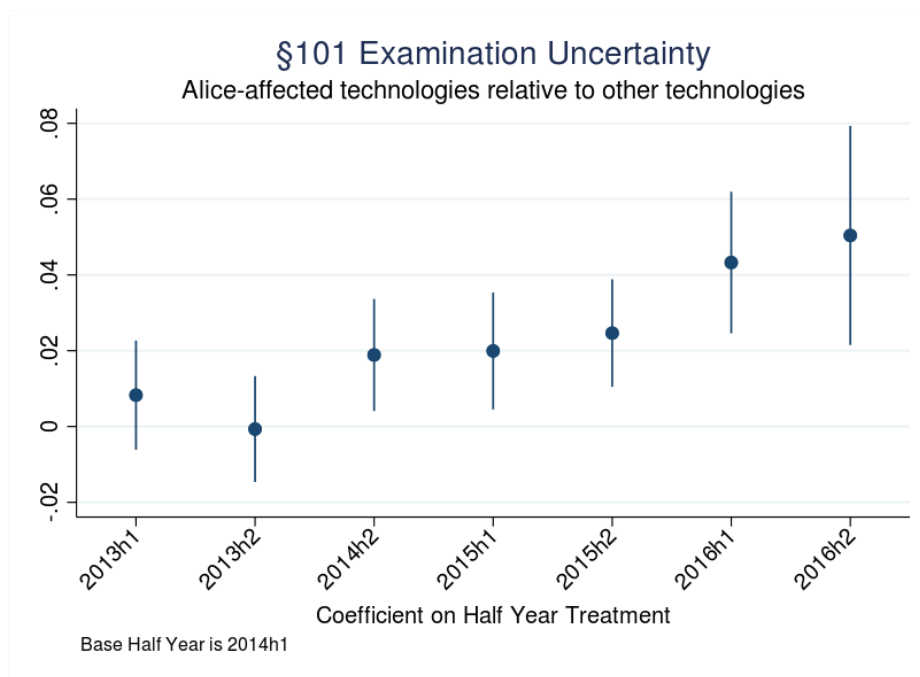
*** p<0.01, ** p<0.05, * p<0.1

In Model 7, the average abstractness of the patent applications in the technology-half years is added as a co-variate. It is not statistically significant and does not have a direct effect on first action examination uncertainty. Note that the main variable of interest does not change in magnitude or significance. Looking at Model 8 with the triple interaction term, the coefficient estimate is not significant. Note that including the triple interaction term dramatically increased

the standard error for *Alice* x PatGrp and produced a statistically insignificant result. That does not mean that our main variable no longer holds, rather it reflects multicollinearity and indicates that Model 3 is not appropriate. We tested the significance of the overall effect using the mean of the abstractness co-variate, $(\beta_1 + \beta_3 * .0234711)$. It is significant and the main effect remains at 0.0272.

Also of interest are the variance controls in the regressions. As expected, the average number of applications per examiner reduced the variation in Section 101 determinations across examiners. Also as expected, average examiner variance for their own Section 101 determinations (i.e., within-examiner) is associated with greater first action examination uncertainty. The variance of abstractness from incoming applications is insignificant, which is consistent with the findings for the other application language abstractness co-variates.

Figure 2 is a coefficient plot from eq. (6) that shows the interactions between the *Alice*-affected technologies group and the half-year time indicators. The base period is the half year directly preceding *Alice* (the first half of 2014). There is a substantial and persistent increase immediately after *Alice*. Further, the pre-trend assumption is satisfied throughout the pre-event period (none of the pre-*Alice* coefficients are statistically different from zero). Using this model and relative to the other technologies, first action examination uncertainty for *Alice*-affected technologies increased by 26% by December 2015.⁴



⁴ This computation is identical to that described in the percentage change footnote in the previous section.

Figure 2: Examination uncertainty in *Alice*-affected technologies relative to other technologies. This figure shows the coefficients for the treatment group interacted with each half year time period in model (8). Importantly, the pre-trend is statistically indistinguishable from zero, where the base period is the first half of 2014. Further, examiner uncertainty increases in the *Alice*-affected technologies post *Alice*.

Berkheimer memo and 2019 PEG effects on first action examination uncertainty

Table 5 shows the regression results for the effects of the Berkheimer memorandum and the 2019 PEG on first action examination uncertainty. For comparison, Model 6 in column (2) presents the results previously discussed for the *Alice* decision from Table 4. Model 9, in the next column, shows that the Berkheimer memorandum had a negative, but statistically insignificant effect on first action examination uncertainty. This means that Berkheimer memorandum did not have a statistically distinct effect on *Alice*-affected technologies relative to other technologies.

Model 10 in Table 5 shows that the 2019 PEG reduced first action examination uncertainty. The coefficient is negative and significant at the 1% level. Using a model with time period interactions with the *Alice*-affected technologies (similar to the model in equation (6)), relative to the other technologies, first action examination uncertainty decreased by 44% in the 12 months following the issuance of the 2019 PEG.

Table 5: Regression Results for the Berkheimer memo and 2019 PEG on Examination Uncertainty

VARIABLES	Model 6	Model 9	Model 10
Alice x PatGrp	0.0272*** (0.00453)		
Berkheimer x PatGrp		-0.00742 (0.00521)	
2019 PEG x PatGrp			-0.0369*** (0.00473)
Avg. number of independent claims	-0.00409 (0.00465)		
Avg. word count in claims	-9.23e-05 (8.29e-05)		
Avg. examiner variance	0.144* (0.0826)	0.0995 (0.0710)	0.139** (0.0686)
Number of examiners	-0.000105*** (3.32e-05)	-7.45e-05** (3.25e-05)	-7.28e-05*** (2.23e-05)
Number of applications	8.36e-06** (3.77e-06)	7.40e-06 (6.01e-06)	1.08e-06 (3.49e-06)
Avg. applications per examiner	-0.00350*** (0.000780)	-0.00185*** (0.000596)	-0.000616** (0.000287)
Variance in abstractness	-43.60 (54.07)		
Constant	0.0791*** (0.0179)	0.0509*** (0.00425)	0.0368*** (0.00301)
Observations	3,030	2,361	2,747
R-squared	0.551	0.645	0.605
Time FE	Half Year	Half Year	Half Year
Technology FE	USPC	USPC	USPC

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Robustness checks for the impact of *Alice*

Assessing dataset restrictions

The analysis focused on the first action stage of patent examination and restricted the datasets to only those patent applications filed before the events of interest in order to minimize the influence of other decision makers on our results. By limiting the datasets to applications filed before the *Alice* decision, we induced a mechanical drop in the number of observations over time – as examiners completed first office actions the remaining pool of applications awaiting a first office action shrinks. This is potentially problematic for our estimates of first action examination uncertainty since fewer applications with first actions will artificially increase the variance across examiners in a way that does not reflect variation in decision-making. We added several co-variables to help capture part of this effect such as the number of applications, the number of applications per examiner, and a flexible time trend using indicator variables.

Figure 3 illustrates the implications of the dataset restriction. It shows the percentage of first office action non-final and final rejections in our dataset relative to the total number of first action rejections in each period. The left censoring reflects the construction of the USPTO Office Action Research Dataset for Patents dataset. It includes patent applications with series numbers that start with 12, 13, 14 and 15, rather including all applications in a fixed period time (see <https://www.uspto.gov/patents-application-process/checking-application-status/search-application>). In figure 3, the vertical bar shows the month of the *Alice* decision, June 2014. As expected, the percentage of first action rejections in our dataset starts to decrease immediately. By December 2015, the percentage had decreased to approximately 45%.

To test the sensitivity of our results on examination uncertainty shown in Table 4, we re-estimated the regression models using the time window of 2013 through 2015. Table 6, Models 11 and 12, shows that the effect of the *Alice* decision on first action examination uncertainty is slightly smaller in magnitude, but statistically significant at the 1% level. Unlike in Table 4, the triple interaction term in Model 13 is positive and statistically significant at the 5% level. This result suggests that the abstract language content of patent applications further increased first action examination uncertainty. The total treatment effect is .018 ($\beta_1 + \beta_3 * .0234302$) and is significant at the 1% level.

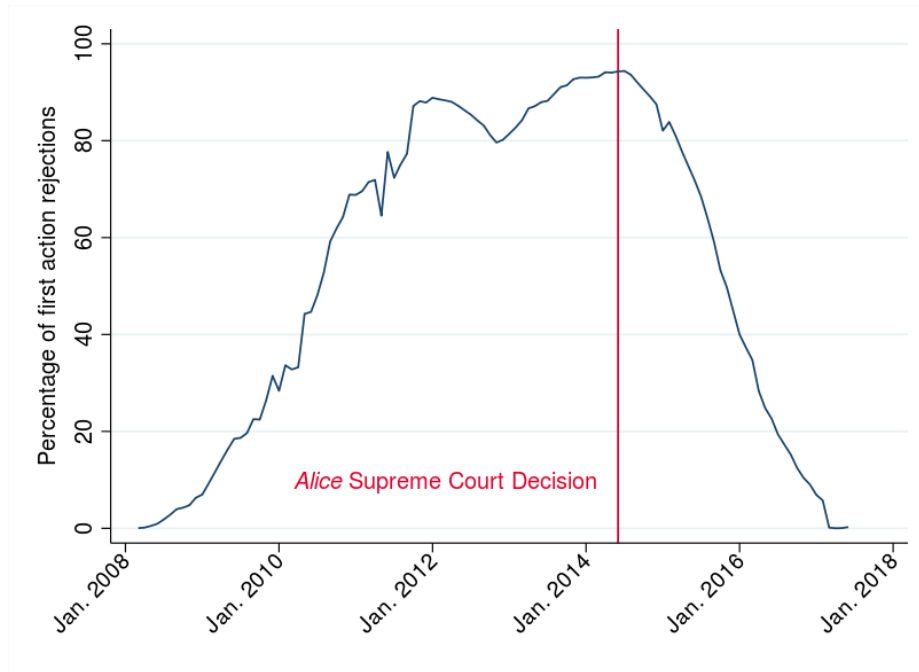


Figure 3: Percentage of first office action rejections in dataset

Table 6: Regression Results for the Test of Dataset Restriction on Applications

VARIABLES	Model 11	Model 12	Model 13
Avg. abstractness		-0.904 (0.869)	-0.907 (0.870)
<i>Alice</i> x PatGrp	0.0185*** (0.00363)	0.0185*** (0.00364)	-0.0165 (0.0154)
<i>Alice</i> x PatGrp x Avg. abstractness			1.488** (0.629)
Avg. number of independent claims	-0.00509 (0.00584)	-0.00483 (0.00575)	-0.00478 (0.00575)
Avg. word count in claims	-4.23e-05 (9.40e-05)	-3.15e-05 (9.42e-05)	-2.89e-05 (9.43e-05)
Avg. examiner variance	0.347*** (0.0664)	0.351*** (0.0667)	0.358*** (0.0667)
Number of examiners	7.86e-05** (3.88e-05)	7.87e-05** (3.87e-05)	7.06e-05* (3.80e-05)
Number of applications	-3.72e-06 (3.57e-06)	-3.74e-06 (3.56e-06)	-3.55e-06 (3.42e-06)
Avg. applications per examiner	-0.000935* (0.000486)	-0.000949* (0.000484)	-0.000917* (0.000483)
Variance in abstractness	-16.58 (39.39)	-7.877 (38.90)	-7.518 (38.90)
Constant	0.0480** (0.0205)	0.0607** (0.0259)	0.0605** (0.0259)
Observations	2,358	2,358	2,358
R-squared	0.633	0.634	0.634
Time FE	Half Year	Half Year	Half Year
Technology FE	USPC	USPC	USPC

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Assessing the Influence of Supreme Court Case *Bilski v. Kappos*

As discussed in the data construction section, the time dimension for this analysis started in 2013 instead of 2011. This choice was made to minimize any impact from the *Bilski* decision on the pre-event trends for *Alice*-affected technologies and other technologies. Rather than limiting the pre-event period, this robustness check allows for a linear time trend difference between the *Alice*-affected and other technologies to account for the impact of *Bilski*. Specifically, we re-run our specifications with the term $t * PatGroup_{it}$ on the sample (2011 – 2016), and assess the appropriateness of a linear time trend using equations (4) and (6).

The regression results, not shown but available upon request, indicate the *Alice* decision had a larger effect on the likelihood of receiving a first office action with a Section 101 rejection, 0.0596 and significant at the 1% level. For first action examination uncertainty, the magnitude was almost the same, -0.0241 and significant at the 1% level. The main results hold up to expanding the dataset time period to 2011, the earliest dataset date permitted by available data.

Potential effects of *Alice* on first office action 102/103/112 rejections

In the final robustness check, we assess the impact of *Alice* on the other main rejection types issued by examiners in the first office action, including Section 102 (novelty), Section 103 (non-obviousness) and Section 112 (clarity and enablement). *Alice* should not impact an examiner's use of these other rejection types for two reasons: (1) *Alice* clarified the law on subject matter eligibility (not novelty, non-obviousness or clarity and enablement issues); and (2), examiners are trained to issue all grounds of rejection on the first office action under the USPTO's compact prosecution policy.⁵ The second point implies that examiners shouldn't substitute away from other rejections when issuing more Section 101 rejections after *Alice*.

We estimate equations (4) and (6) with each of the other rejections types. The time period point estimates are provided in Figures 4-9 below. Importantly, none of the figures show a statistically significant discontinuous jump in the rejection rates after *Alice*.

⁵ A description of the USPTO's compact prosecution policy is located at: <https://www.uspto.gov/web/offices/pac/mpep/s2103.html>

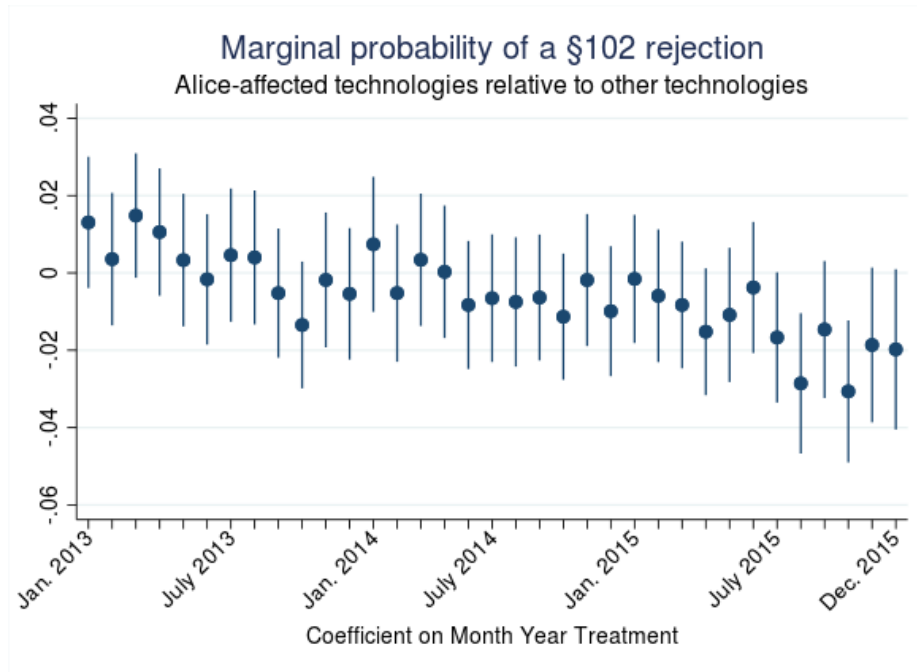


Figure 4: Regression coefficients for Section 102 rejections

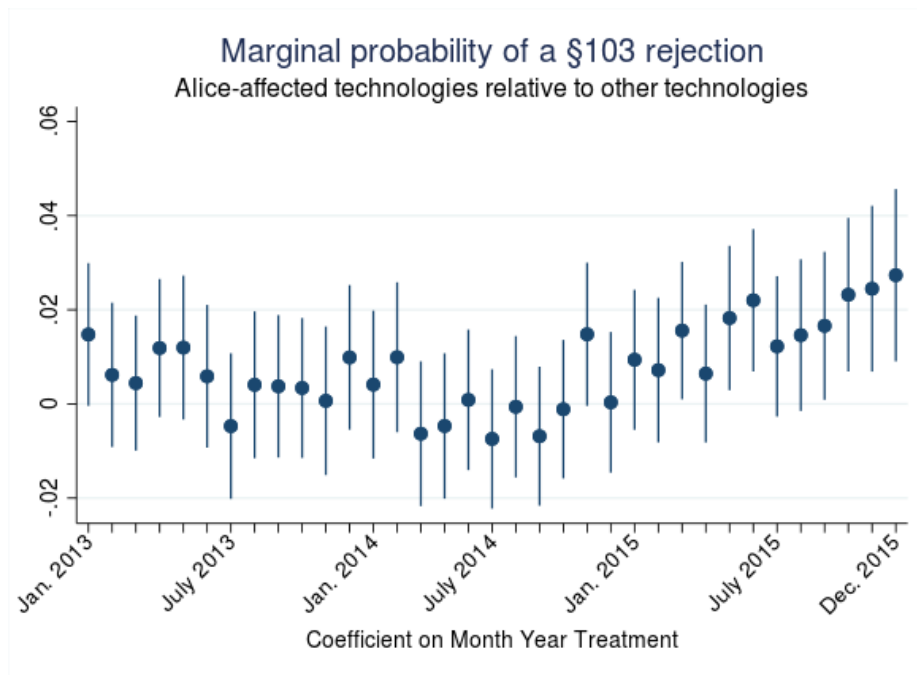


Figure 5: Regression coefficients for Section 103 rejections

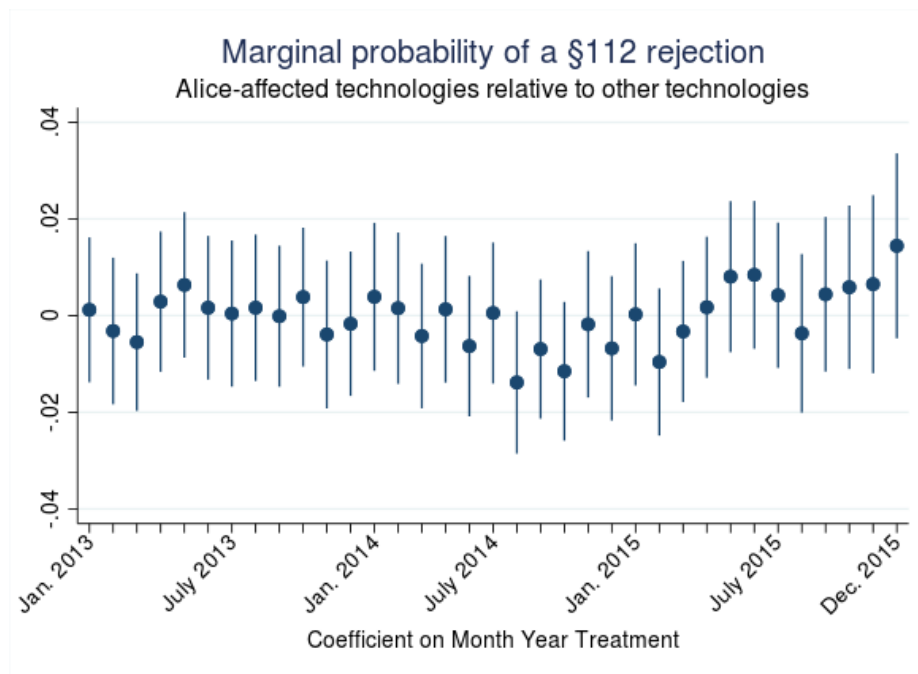


Figure 6: Regression coefficients for Section 112 rejections

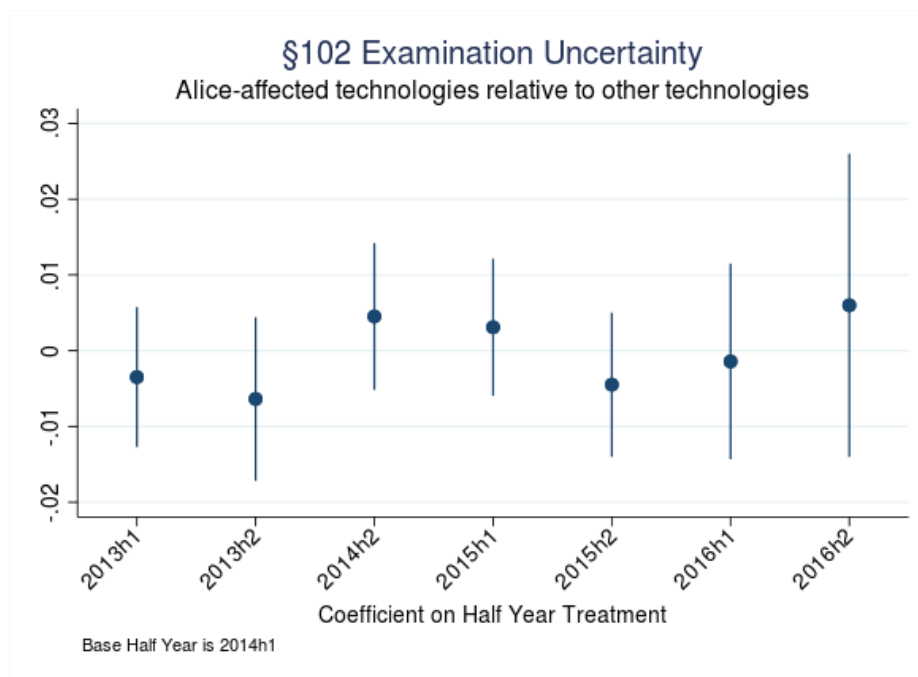


Figure 7: Regression coefficients for Section 102 examination uncertainty

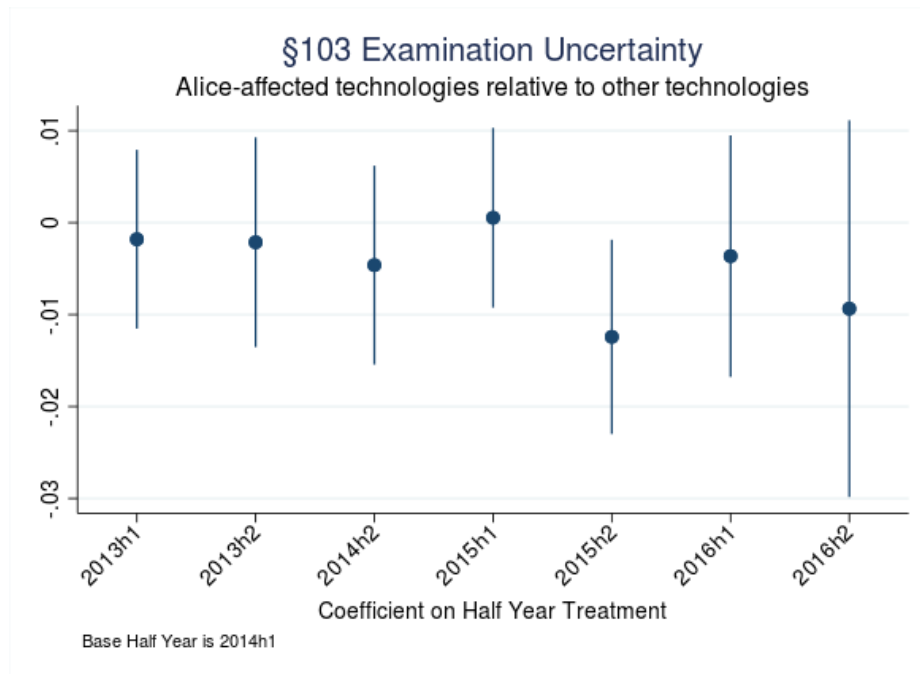


Figure 8: Regression coefficients for Section 103 examination uncertainty

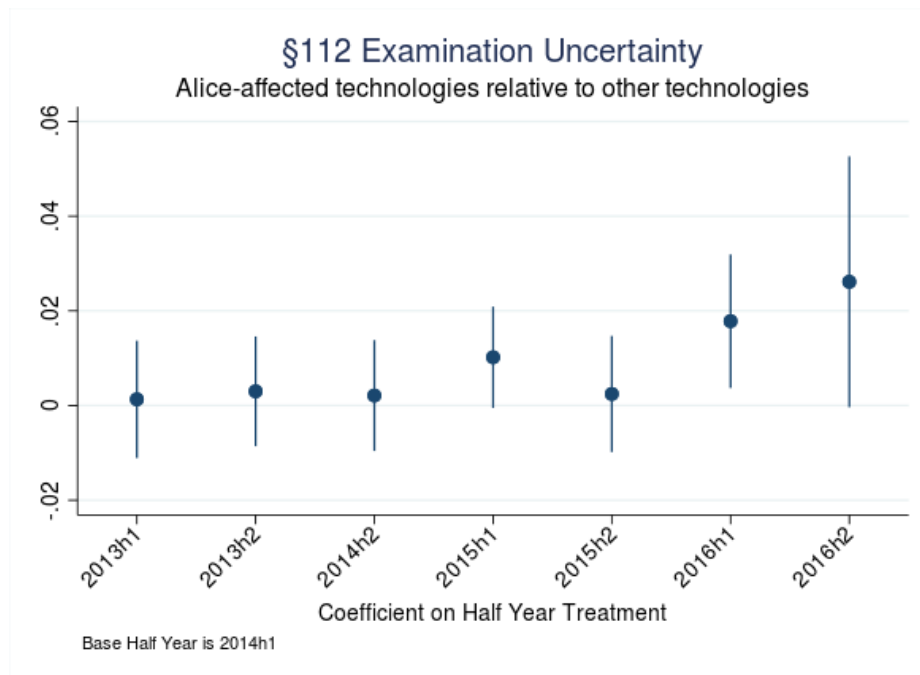


Figure 9: Regression coefficients for Section 112 examination uncertainty